Fiscal Year:	FY 2025	**Task Last Updated:**	FY 10/30/2024
PI Name:	Handakumbura, Pubudu Pinipa Ph.D.		
Project Title:	C4 Photosynthesis in Space (C4Space)		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	99354	**Congressional District:**	4
Comments:	The Battelle Environmental Molecular Science Laboratory (EMSL) is a US Department of Energy (DOE) funded facility and is part of the Pacific Northwest National Laboratory (PNNL).		
Project Type:	Flight	**Solicitation / Funding Source:**	2018 Space Biology (ROSBio) NNH18ZTT001N-FG. App B: Flight and Ground Space Biology Research
Start Date:	01/01/2021	**End Date:**	03/31/2026
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA KSC
Contact Monitor:	Massa, Gioia	**Contact Phone:**	321-861-2938
Contact Email:	gioia.massa@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 03/31/2026 per PI and G. Massa/KSC (Ed., 6/4/25) NOTE: End date changed to 12/31/2024 per PI (Ed., 11/1/24) NOTE: End date changed to 06/30/2024 per PI and G. Massa/KSC (Ed., 6/23/23)		
Key Personnel Changes/Previous PI:	Per the PI, CoInvestigator has left the project (Ed., 11/1/24). The previous Principal Investigator (PI) was Dr. Christer Jansson at the Pacific Northwest National Laboratory (PNNL) from February 2020-December 2021. Dr. Jansson retired at the end of 2020. Dr. Pubudu Handakumbura, who was a CoI (CoInvestigator) on the original grant, took over as PI in January 2021.		
COI Name (Institution):	Hixson, Kim Ph.D. (PNNL / BATTELLE EMSL for the US Department of Energy)		
Grant/Contract No.:	Department of Energy IAA		

Performance Goal No.:	
Performance Goal Text:	
Task Description:	NOTE: This project, (Department of Energy IAA), is a continuation of the investigation titled, "C4 Photosynthesis in Space (C4Space)", with Dr. Pubudu Handakumbura from the Pacific Northwest National Laboratory (PNNL) as the new Principal Investigator (PI). C4 plants like maize (Zea mays) and sorghum (Sorghum bicolor) have a more efficient photosynthesis than C3 plants such as wheat (Triticum aestivum) and rice (Oryza sativa) due to a CO2-concentrating mechanism (CCM). How this CCM and the performance of C4 plants are impacted by space travel is unknown. We propose to compare the impact of space-station conditions on C3 and C4 metabolism using Brachypodium (Brachypodium distachyon) and Setaria (Setaria viridis) as model systems for C3 and C4 plants, respectively, and develop models that describe the molecular mechanisms for how C3 and C4 metabolisms are reprogrammed in the space environment compared to Earth. The obtained information would provide fundamental knowledge about C3 and C4 metabolism in space and could also be leveraged for evaluating the potential for growing small-stature cereal and vegetable C4 crops like foxtail millet (Setaria italica) and Amaranthus sp. for biogenerative support in future space explorations.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The research provides fundamental understanding of plant biochemistry in space environments as well as an assessment of the suitability in utilizing C4 crops in biogenerative life support systems in future space explorations.
Task Progress:	2024: The Advanced Plant Experiments (APEX) 09 team successfully carried out the Experiment Verification Test (EVT) and was approved to set up the Flight experiment to launch to the International Space Station (ISS) after passing the Flight Readiness Review in February of 2024. The APEX 09 experiment was launched to the ISS in NG21 on August 4th, 2024. On-orbit Brachypodium seeds germinated as expected; however, the Setaria seeds failed to germinate. Despite the extensive post-flight testing, the cause of poor Setaria seed germination remains elusive. The 32-day-long Brachypodium growth experiment was completed on orbit and Earth. The ISS Brachypodium samples are anticipated to return to Earth on SpaceX-31 in late November 2024. 2023: A second Science Verification Test (SVT) was conducted at NASA Kennedy Space Center (KSC) on May 17th. Dr. Handakumbura, Project Principal Investigator (PI) carried sterilized Brachypodium and Setaria seeds to the Kennedy Space Center and prepared 24 plant growth systems (PGS) for the second Science Verification Test (SVT). Brachypodium and Setaria were successfully grown in the PGS for 32 days in the NASA Vegetable Production System (Veggie growth system) under a simulated International Space Station environment for CO2. Two harvests were performed during the growth period by the NASA team at KSC. Dr. Handakumbura assessed the SVT2 samples against the established success criteria at PNNL. The PI team at PNNL tested alternative seed sterilization techniques to optimize Setaria plant performance further. After a successful report to the NASA program managers, the C4 Space project was approved to move forward to conduct an Experimental Verification Test (EVT). Similar to SVT2, Dr. Handakumbura hand-carried sterilized Brachypodium and Setaria seeds and assembled 24 PGS at KSC on September 20th. Brachypodium and Setaria plants were successfully grown for the 32-day growth period at KSC. Both Brachypodium and Setaria had robust growth under the ISS CO2 levels. Plant samples were harvested from the 24 PGS at two developmentally distinct time points and immediately frozen and stored in a -80C freezer until return to the PI. Samples were shipped to PNNL at end of October and the PI team evaluated the samples against the established success criteria. 2022: C4 Space project plant growth system (PGS) hardware configurations were completed in 2022. The growth of Brachypodium distachyon and Setaria viridis was tested on two different growth setups using Oasis foam and Arcillite with the modified hardware. Seeds were directly planted on form and germination paper wicks were used to hold Brachypodium and Setaria seeds with Arcillite. Based on the plant growth performance, Arcillite was selected as the growth medium. Seed placement within the germination paper was tested and optimized. Watering volumes and watering frequencies for the two plant species were also optimized during the year 2022. A 3D-printed germination paper guide was developed by the PI to hold the seed-glued germination paper wicks in place during packing and assembly of the PGSs. Based on all the above tests, a 32-day-long experimental workflow was developed for Brachypodium and Setaria. This workflow was tested multiple times by the PI and the team and a revised success criteria was established for the C4 Space project. Pre-science verification test (SVT) Science Readiness review was conducted by the NASA team in September 2022 and the PI team was approved to conduct a SVT at Kennedy Space Center (KSC) using the Veggie growth system mimicking International Space Station (ISS) environmental conditions. The SVT1 was started in mid-October 2022 and all samples were harvested and returned to PI by November 30th, 2022. SVT was a complete success for Brachypodium; however, Setaria failed to germinate during SVT1.
Bibliography Type:	Description: (Last Updated:)